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Question 6 **(25 marks)**

The general term of an arithmetic sequence is $T_n = 15 - 2n$, where $n \in \mathbb{N}$.

(a) **(i)** Write down the first three terms of the sequence.

$T_1 = 15 - 2(1) = 13$

$T_2 = 15 - 2(2) = 11$

$T_3 = 15 - 2(3) = 9$

$13, 11, 9$

(ii) Find the first negative term of the sequence.

$T_8 = 15 - 2(8) = -1$

$T_8 = -1$

(b) **(i)** Find $S_n = T_1 + T_2 + \cdots + T_n$, the sum of the first n terms of the series, in terms of n.

$S_n = \frac{n}{2}\left(2(13) + (n-1)2\right)$

(ii) Find the value of n for which the sum of the first n terms of the series is 0.

$\frac{n}{2}\left(2(13) + (n-1)2\right) = 0$

$\frac{n}{2}(26 + 2n - 2) = 0$

$\frac{n}{2}(24 + 2n) = 0$

$24 + 2n = 0$

$2n = -24$

$n = -12$

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